Hyeonbin Han

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EDUCATION

Hanyang University ERICA **Expected Graduation Date:** February 2027
GPA: 3.51/4.5

Department of Mathematical Data Science

RESEARCH EXPERIENCE

CMES Robotics, Seoul, Korea **Mar 2026 - Aug 2026**

Junior Research Assistant, Robot Intelligence Team (Humanoid TF)

- Built and standardized teleoperation and data-to-deployment workflows for UR5 manipulation using GELLO, LeRobot, ACT, and containerized robot-learning software.
- Developed a FastAPI and Next.js GELLO WebUI that unified robot connection, demonstration recording, dataset review, live camera monitoring, and learned-policy deployment.
- Designed real-world ACT experiments for dynamic conveyor assembly, sequential manipulation, and contact-rich packing; diagnosed the effects of camera coverage, action horizons, demonstration composition, and checkpoint selection.
- Built a Cyclo Lab and Zenoh simulation pipeline for residual RL, collected 30 demonstrations, trained an ACT base policy, and defined a controlled ACT-versus-ACT+PPO evaluation protocol; residual training remains in progress.

Intelligence & Computation Lab, Hanyang University ERICA **Oct 2023 - Jan 2026**

Undergraduate Researcher

- Maintained GPU training infrastructure and resource allocation for deep-learning experiments.
- Developed preprocessing pipelines for heterogeneous healthcare and mobile-addiction datasets.

SELECTED RESEARCH PROJECTS

Residual RL for Robot Manipulation **Aug 2026 - Present**

- Formulated a bounded residual controller that combines a frozen ACT action with a PPO correction policy for simulated FFW-SG2 pick-and-place.
- Defined reproducible evaluation across shared seeds using task success, grasp/place success, collision and drop counts, episode return, and residual-action norms; comparative results are pending.
- Built the Cyclo Lab environment and Zenoh-controlled collection path with wrist-camera observations; collected 30 demonstrations and trained the initial ACT baseline.

Single-Arm Imitation Learning for Dynamic Assembly **Jun 2026 - Aug 2026**

- Trained and deployed ACT policies on a UR5 for moving-conveyor gear assembly using staged GELLO demonstrations; identified camera coverage and action-horizon settings as major determinants of grasp behavior.
- Evaluated targeted data collection across layouts, recovery behavior, object colors, and packing variants; observed 90%, 80%, and 50% success across three bundle conditions in 10-trial evaluations.
- Expanded the conveyor dataset from 60 to 110 episodes, reduced the initial ACT chunk size from 50 to 25, and recollected 50 episodes after correcting a camera blind spot.

SELECTED RESEARCH PROJECTS (CONT.)

GELLO WebUI for Robot Learning Workflows — Mar 2026 - Aug 2026

- Built a FastAPI/Next.js operational layer integrating UR5 and GELLO teleoperation, live camera streaming, LeRobot recording, and ACT checkpoint deployment.
- Implemented state-managed practice and recording workflows, explicit hardware and inference error reporting, and a health-checked launcher coordinating backend and frontend processes.

Imitation Learning for Pick-and-Place — Sep 2025 - Dec 2025

- Built an end-to-end LeRobot workflow using 200 teleoperated demonstrations and deployed ACT and Diffusion Policy on the ROBOTIS AI Worker FFW-SG2.
- Tested policy behavior under lighting and environmental variation, motivating subsequent work on failure recovery and residual reinforcement learning.

3D Perception for Robotic Depalletizing — Mar 2025 - Jun 2025

- Developed a YOLOv8-based 3D perception pipeline using depth maps and synthetic Blender data for texture-invariant box detection.
- Computed surface normals, target coordinates, and 6-DoF end-effector orientations and integrated them with an RB5-850 manipulation workflow.

PUBLICATION

Han, Hyeonbin, et al. "Prediction of Closed Quotient During Vocal Phonation Using GRU-Type Neural Network with Audio Signals." *Journal of Information and Communication Convergence Engineering*, 22(2), 145-152, 2024.

CONFERENCE PRESENTATIONS

- "DMD Method for Analyzing Interactions Between Board Game Players," KIICE Fall Conference, 2025. Poster.
- "Deep Learning-Based Algorithm for Analyzing EGG Signals to Predict Closed Quotient Rate," KIIS Spring Conference, 2024. Poster.

SELECTED HONORS

- Excellence Award, Hanyang University ERICA College of Computing Capstone Fair, Jun 2026.
- Excellence Award, Intelligent Robot WE-Meet Project Integrated Competition, Jan 2026.
- Bronze Prize (Robot News President's Award), KIICE Physical AI Challenge, Oct 2025.
- Best Paper Award, KIICE Fall Conference, Oct 2025.

TECHNICAL SKILLS

Robot Learning: PyTorch, Hugging Face LeRobot, ACT, Diffusion Policy, SmolVLA
Robot Platforms & Teleoperation: GELLO, UR5, ROBOTIS FFW-SG2
Systems: Python, Linux, Docker, Git
Perception: YOLOv8, 3D Vision, NumPy
3D Tools: Fusion 360, FreeCAD, 3D Printing